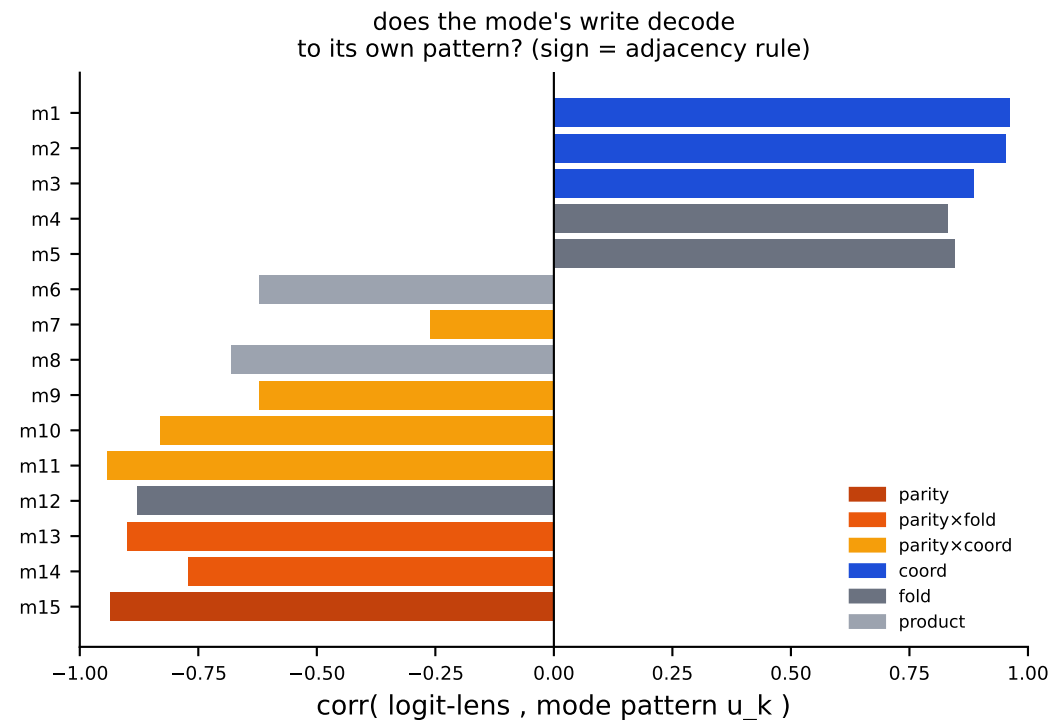
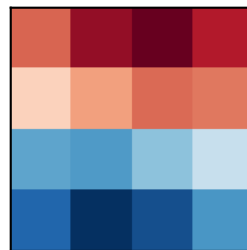


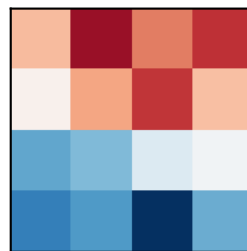
Eigenmode logit lens — Llama, grid (layer 30/32, 2nd-to-last)



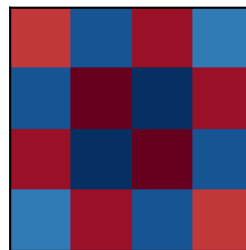
m2 pattern u_k
coord



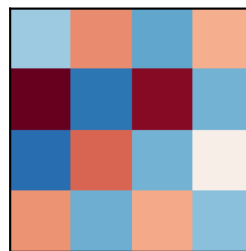
m2 logit-lens
LL·u=+0.95



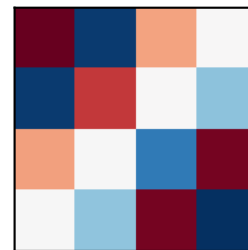
m15 pattern u_k
parity



m15 logit-lens
LL·u=-0.94



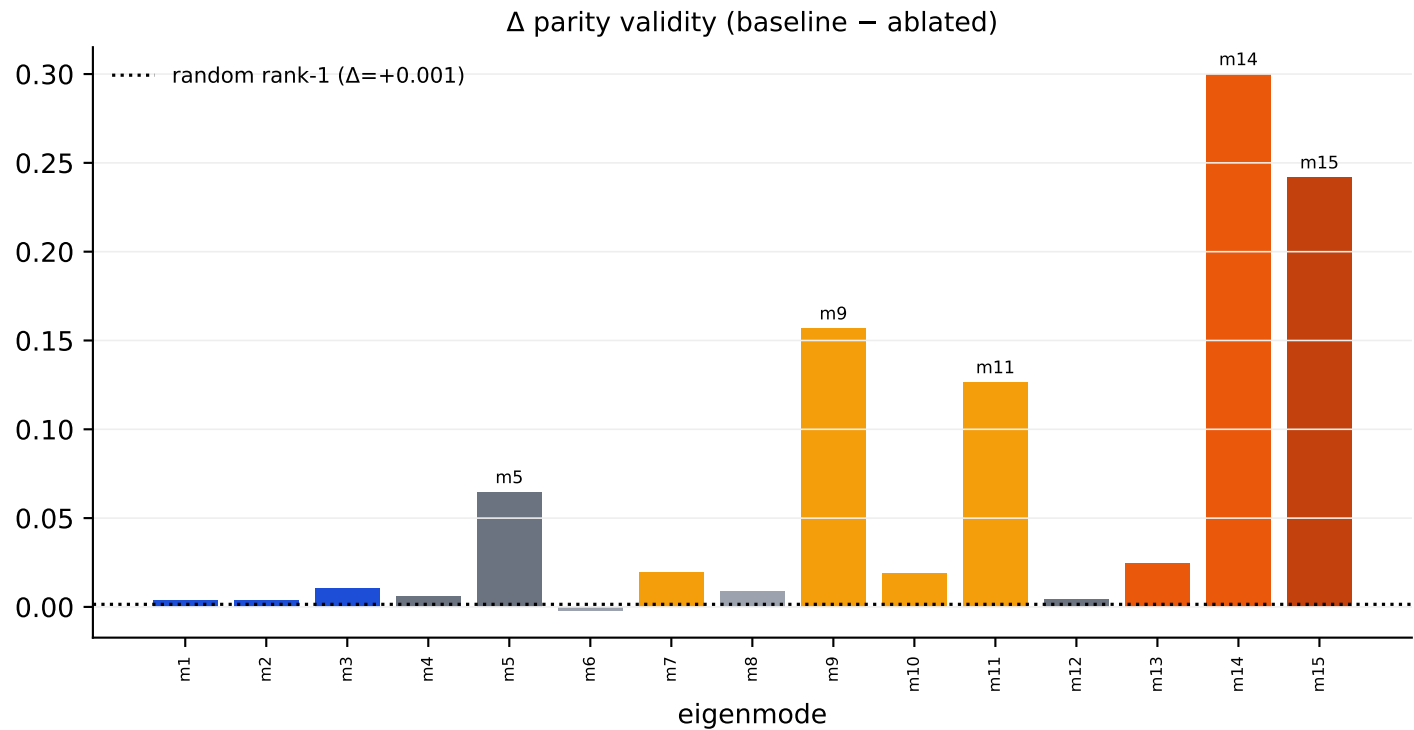
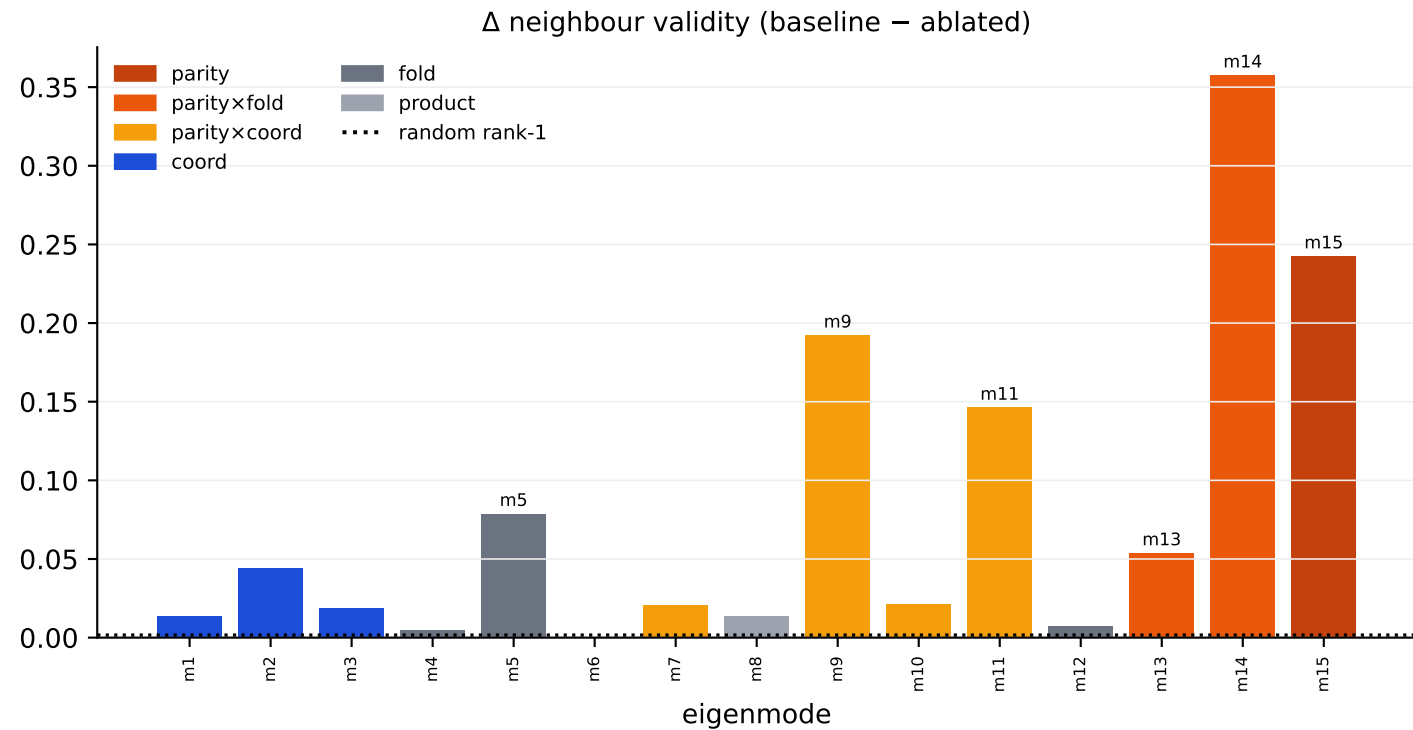
m14 pattern u_k
parityxfold

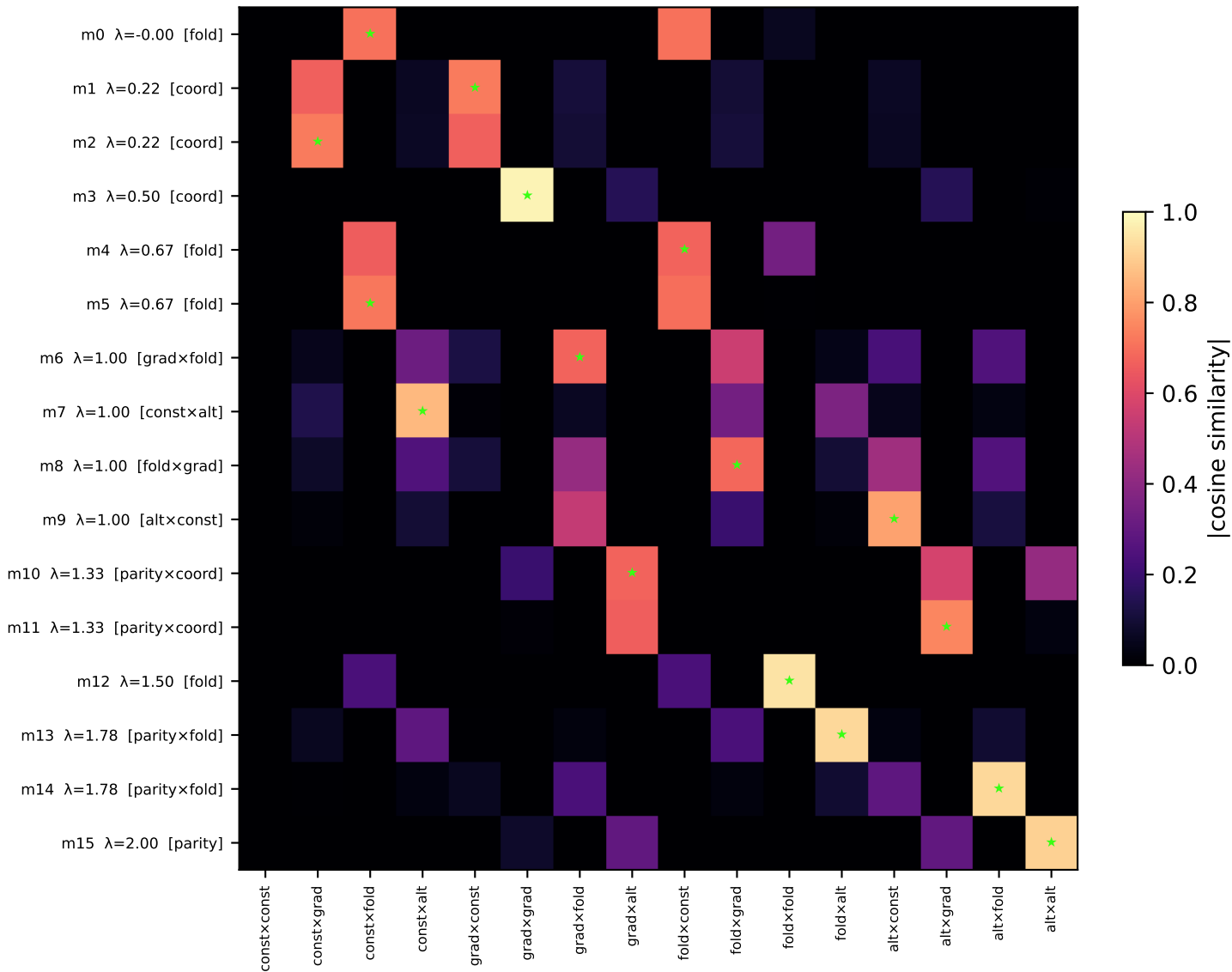


m14 logit-lens
LL·u=-0.77

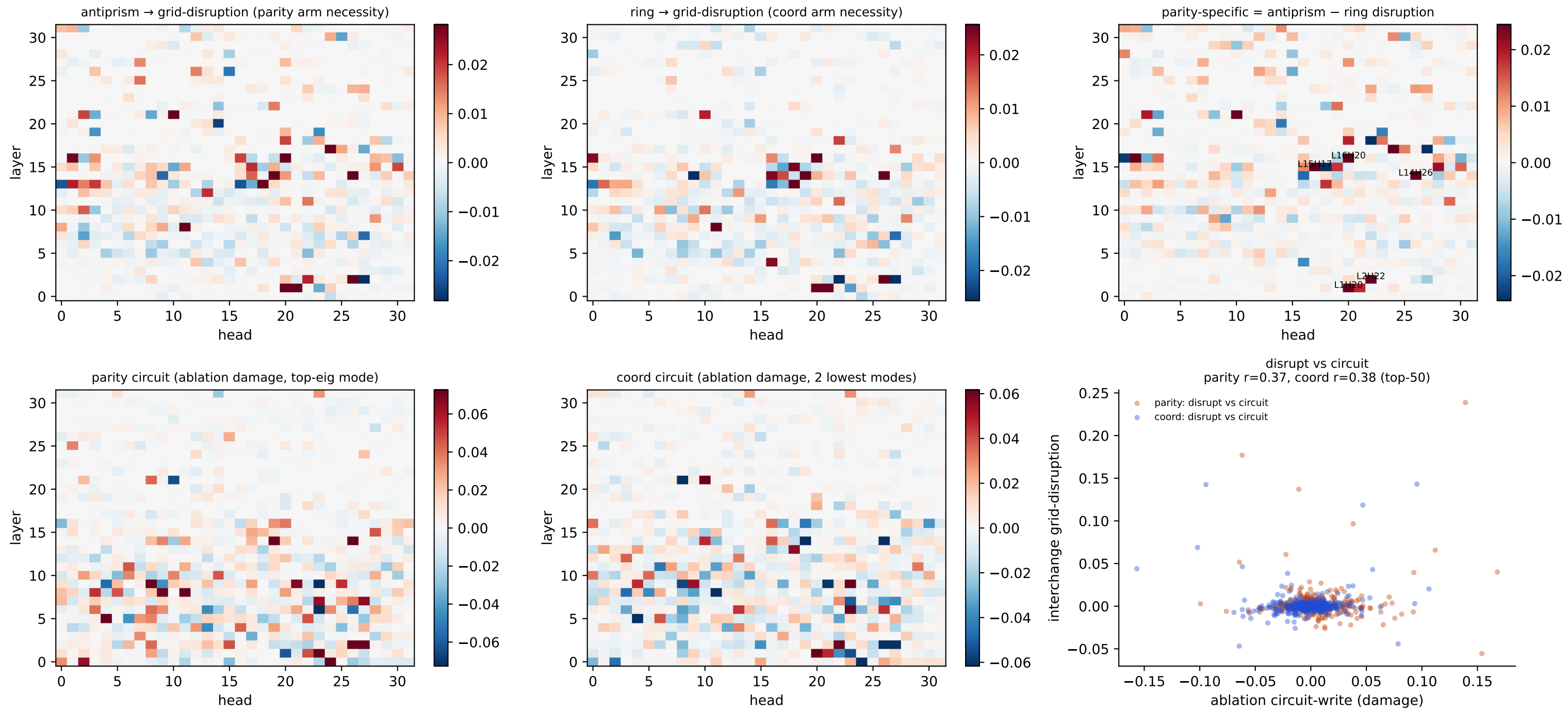


Per-mode single ablation — Llama, grid (baseline nbr=0.99, chance=0.20)

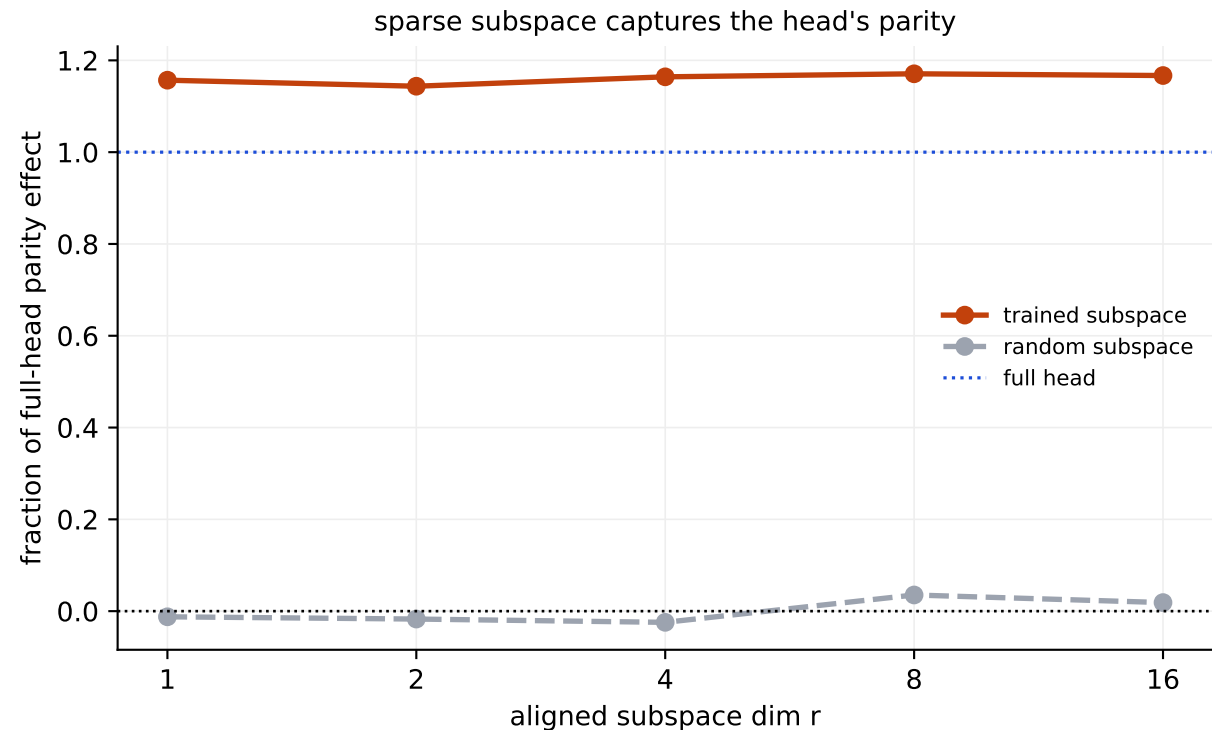
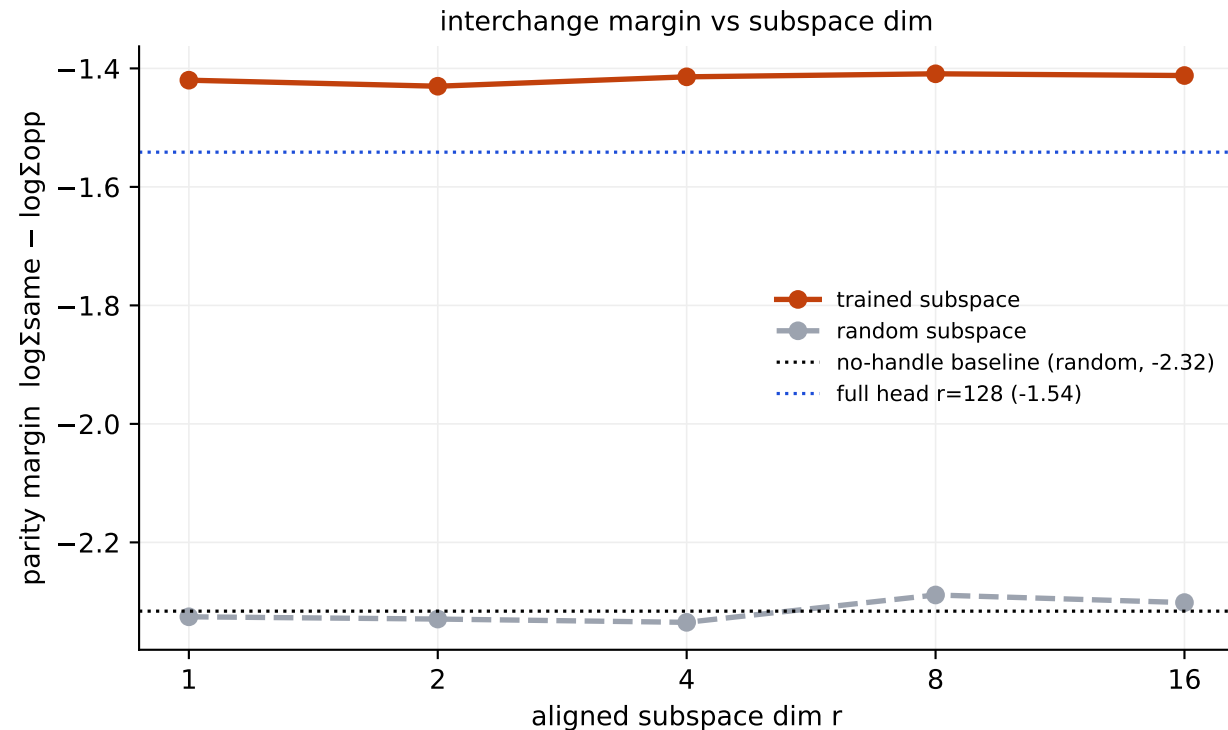


Eigenmode purpose index -- 4x4 grid (Laplacian mode $\times$ Kronecker product)

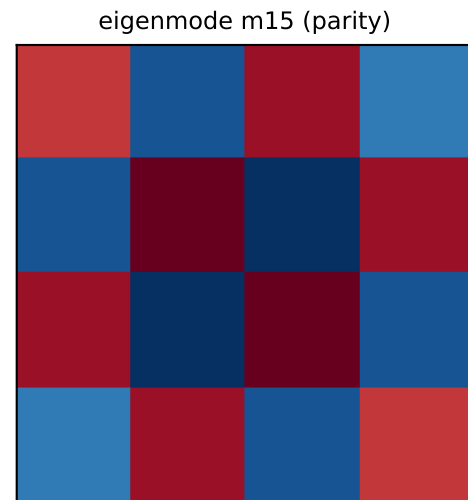
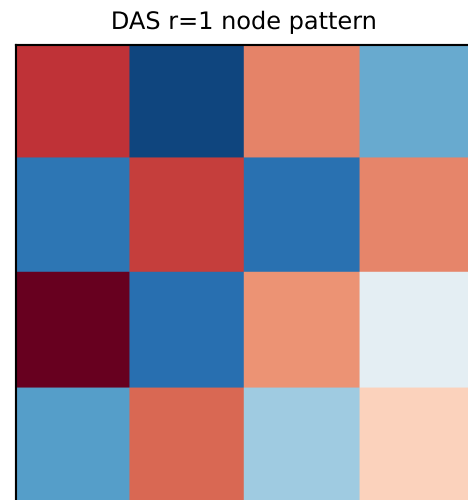
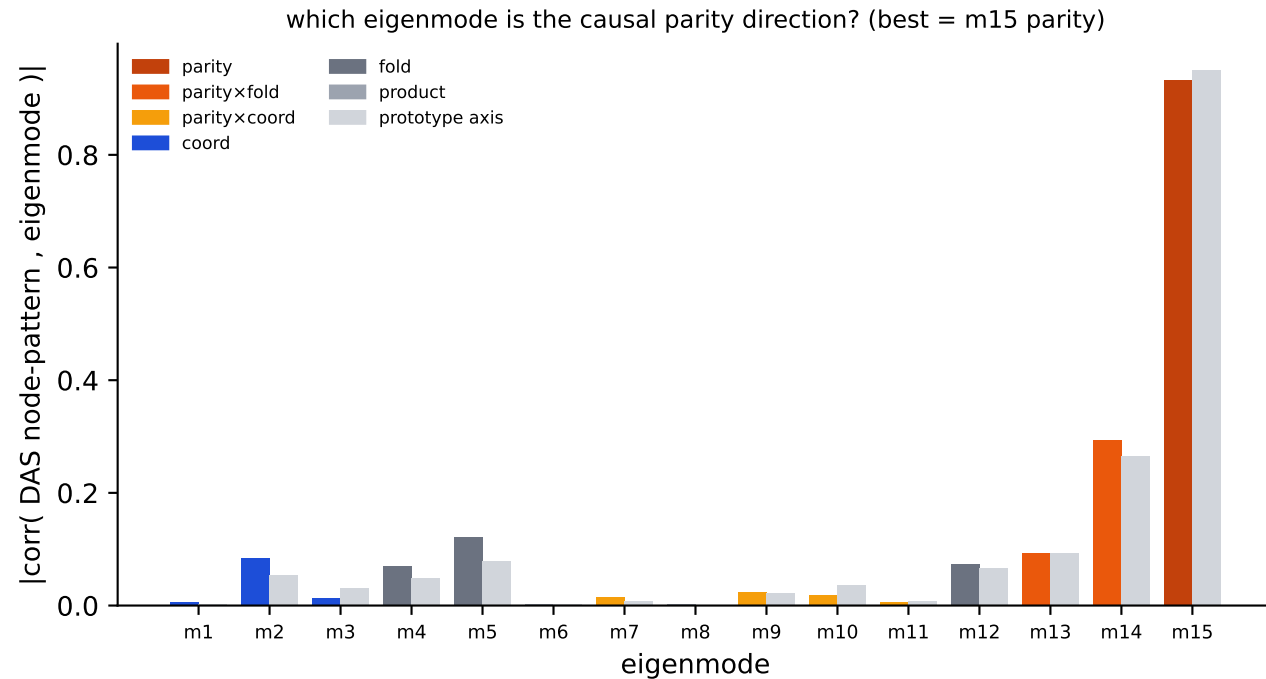
Exp 4 — causal head interchange vs the parity/coord circuit (Llama). Parity-specific top: L14H26, L2H22, L16H20, L15H17



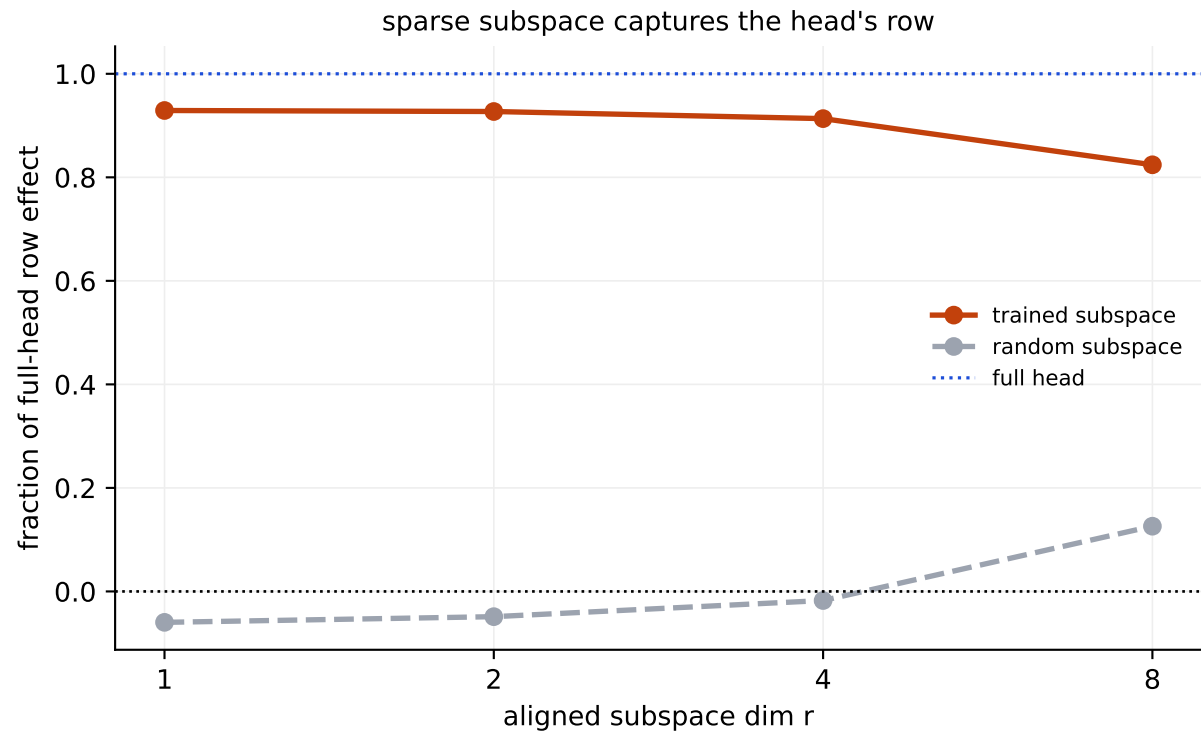
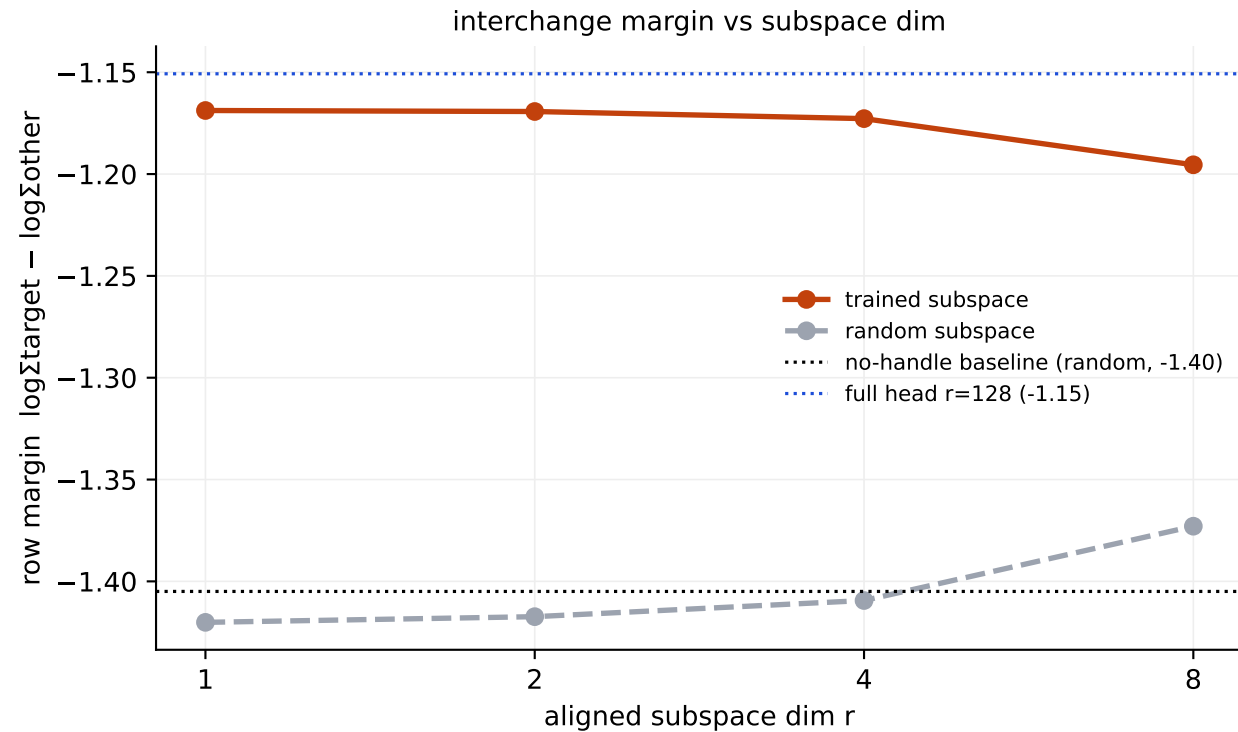
Exp 5 — DAS inside head L14H26 (Llama): is parity in a sparse subspace? (multi-position prototype interchange)



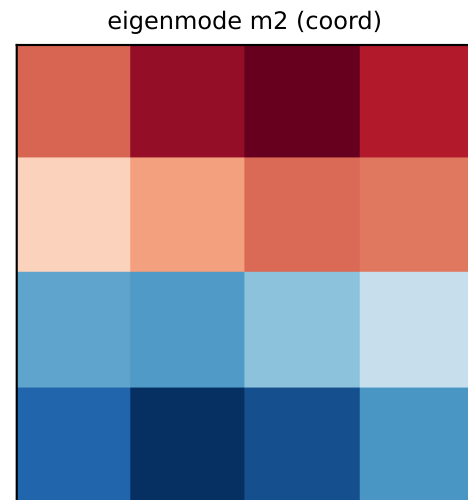
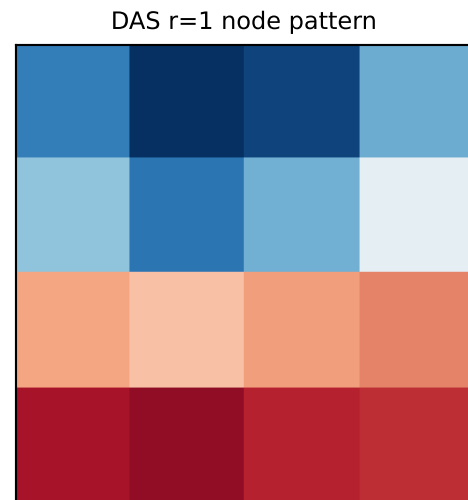
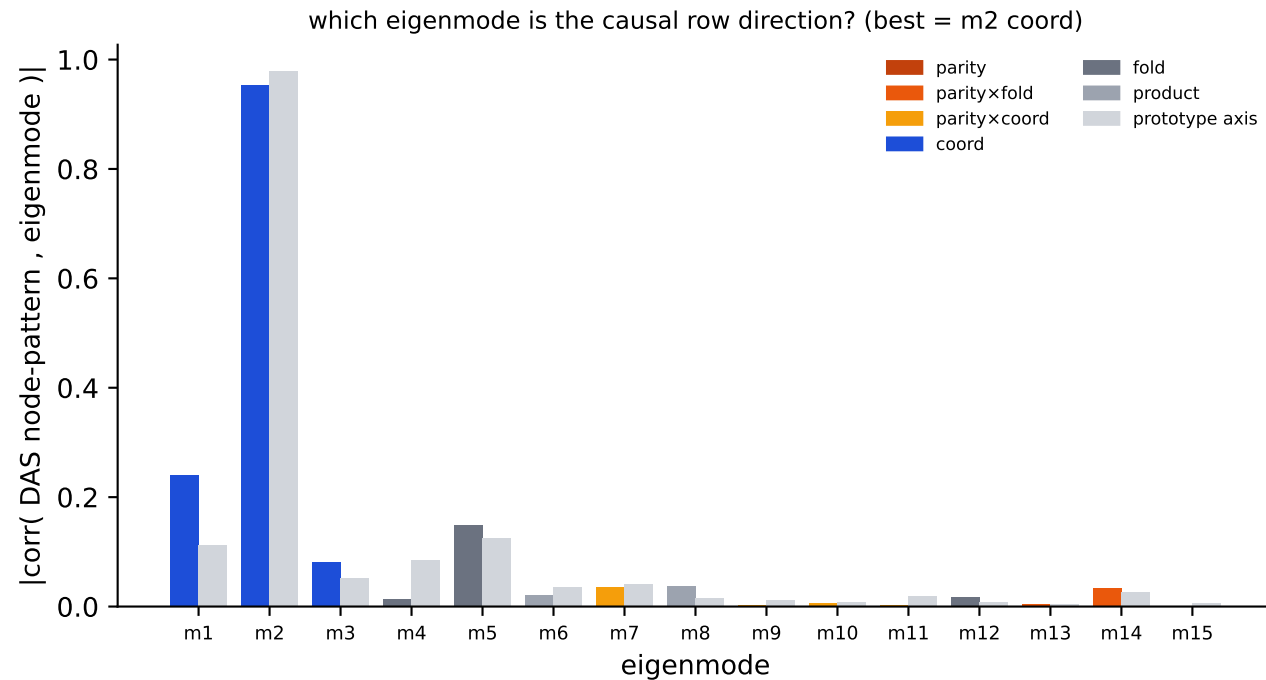
DAS parity direction $\leftrightarrow$ eigenmodes (Llama, L14H26): DAS-prototype $|\cos|=0.95$, best mode m15 $|\text{corr}|=0.93$



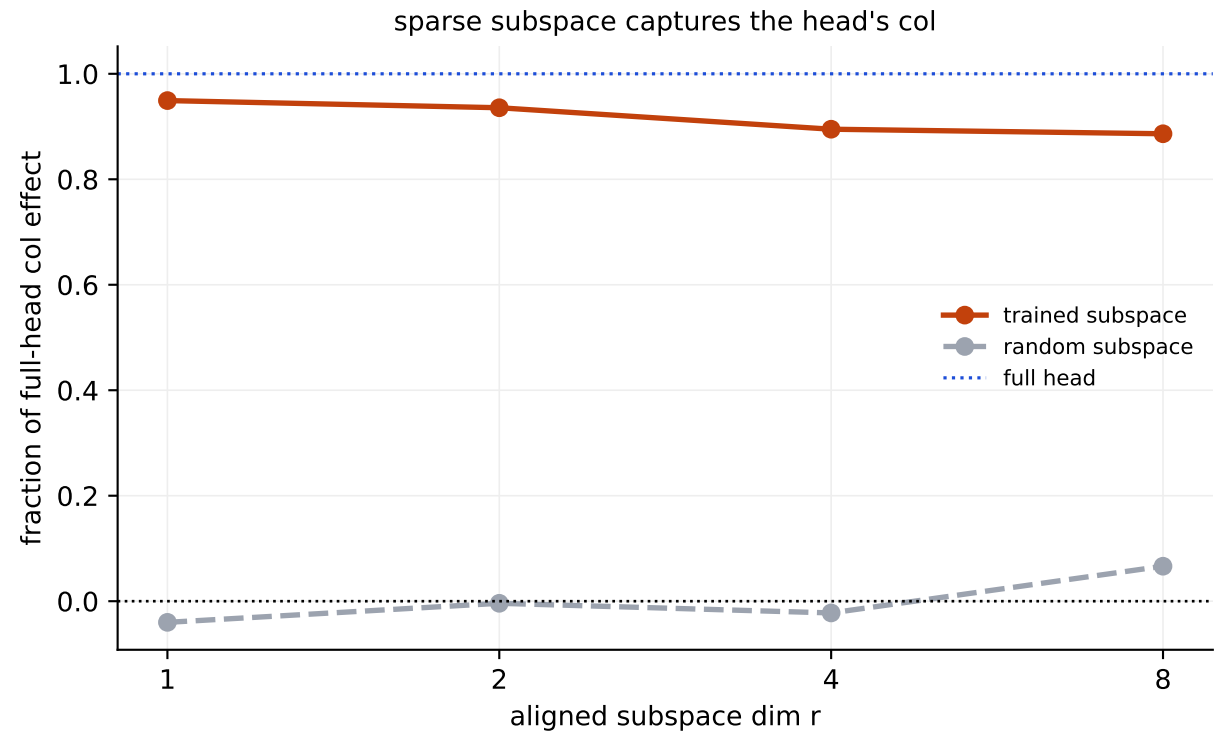
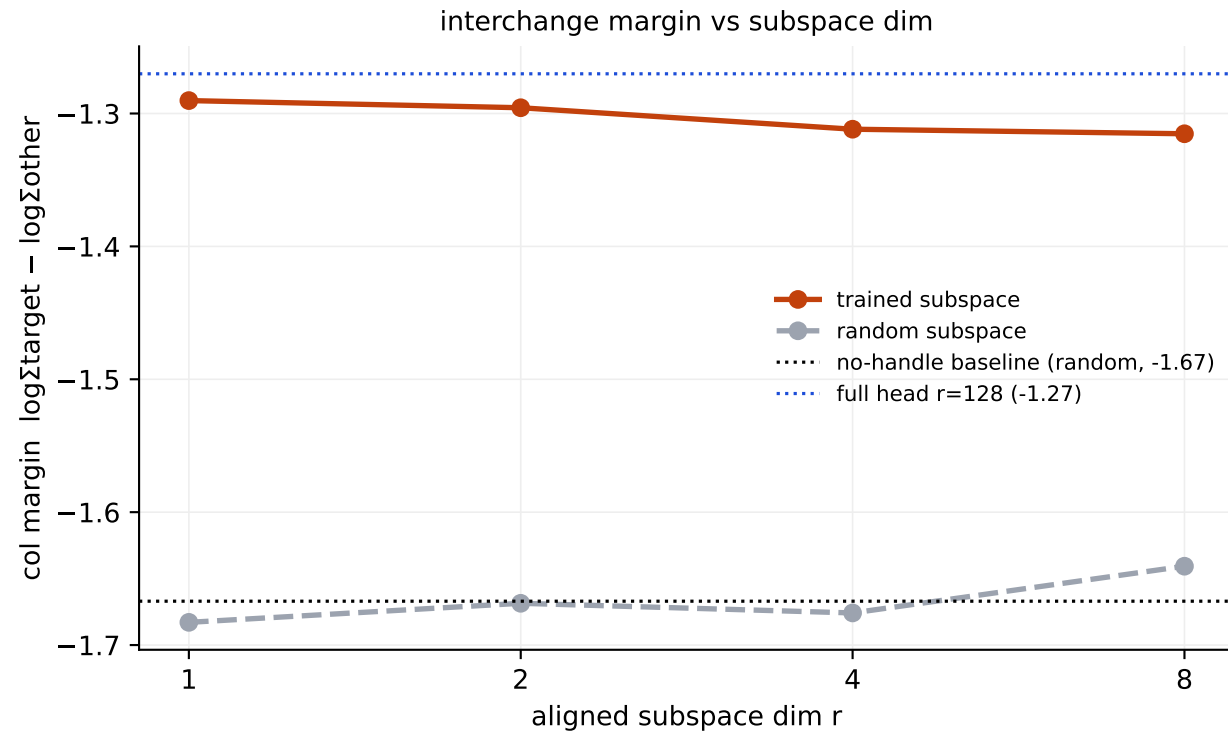
Exp 5 — DAS inside head L21H10 (Llama): is row in a sparse subspace? (multi-position prototype interchange)



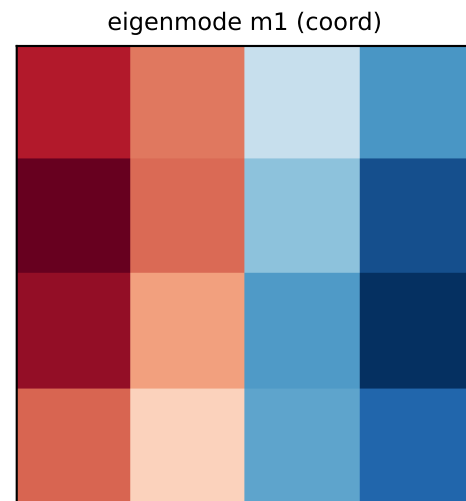
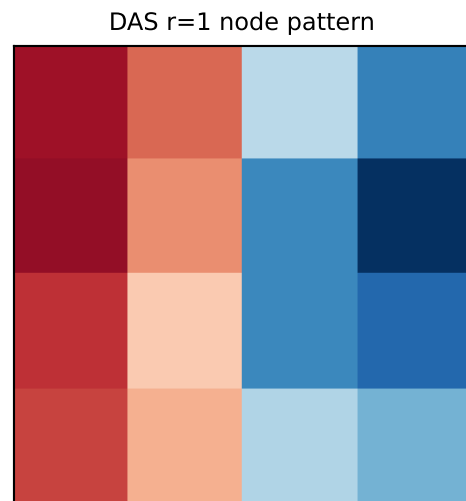
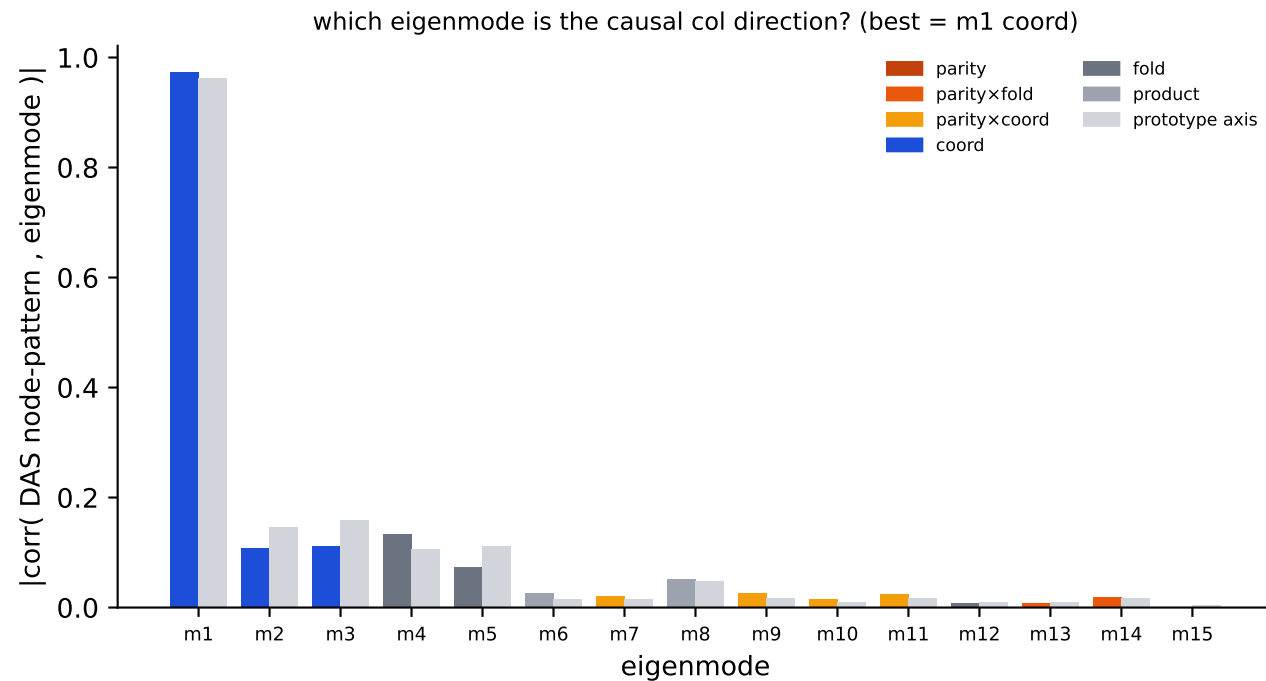
DAS row direction $\leftrightarrow$ eigenmodes (Llama, L21H10): DAS·prototype $|\cos|=0.83$, best mode m2 $|\text{corr}|=0.95$



Exp 5 — DAS inside head L21H10 (Llama): is col in a sparse subspace? (multi-position prototype interchange)



DAS col direction $\leftrightarrow$ eigenmodes (Llama, L21H10): DAS-prototype $|\cos|=0.91$, best mode m1 $|\text{corr}|=0.97$



Exp 6 — the distributed parity circuit: builder → mover → reader across depth (Llama)

